

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 3



Mathematical Preparation

Finite & Infinite Sets

We call two sets A & B **equinumerous** if there is a bijection $f : A \rightarrow B$

A set is **finite** if it is equinumerous with $\{1, 2, \dots, n\}$ for some natural number n .

A set is **infinite** if it is not finite.

A set is **countably infinite** if it is equinumerous with $\mathbb{N}$ the set of natural numbers.

Countably infinite set can be **enumerated** as $\{a_0, a_1, a_2, \dots, a_n, \dots\}$

A set **is countable** if it is finite or countably infinite

A set that is not countable is **uncountable**

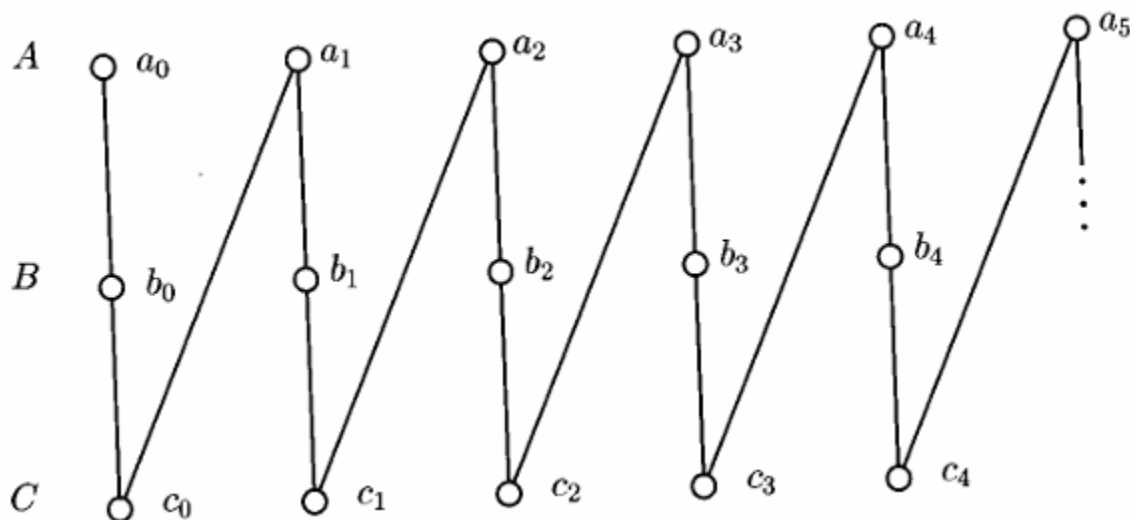
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Important Results

1. Union of any finite number of countably infinite sets is countably infinite

Suppose A , B & C are countably infinite then

$A \cup B \cup C$ is also countably infinite



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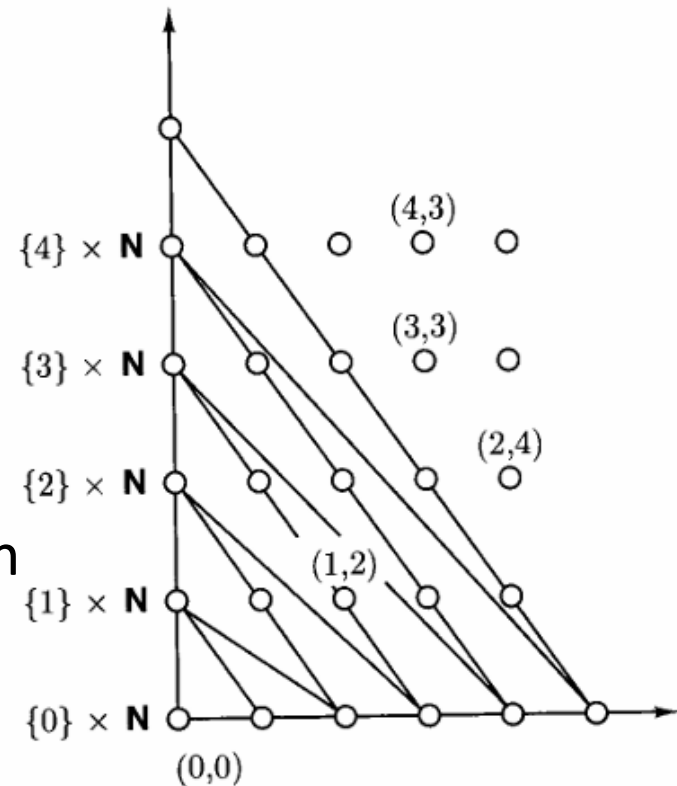
Countably infinite Union of countably infinite sets is countably infinite.

Important Observation

The set $\mathbb{Q}$ of rational numbers is countably infinite

Exercise

1. Find the precise bijection from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$
2. If A & B are countably infinite then $A \times B$ is also countably infinite.
3. What about countably infinite Cartesian product of countably infinite set



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3. The set $2^{\mathbb{N}}$ is uncountable

Proof:

Suppose $2^{\mathbb{N}}$ is countably infinite.

Then we can enumerate all members of $2^{\mathbb{N}}$ as

$$2^{\mathbb{N}} = \{R_0, R_1, R_2, \dots\}$$

Consider the set $D = \{n \in \mathbb{N} : n \notin R_n\}$

Then $D \subseteq \mathbb{N}$ and so $D = R_k$ for some k

We ask the question whether $k \in R_k$

i. If **Yes**, then $k \notin D$ a contradiction as $D = R_k$

ii. If **No**, then $k \in D$ a contradiction as $D = R_k$ so $k \in R_k$

A contradiction

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Closures

Intuitive Idea

Natural numbers N are **closed** under $+$

i.e. for given two natural numbers n, m we always have that $n + m \in N$

Natural numbers N are not closed under subtraction –

i.e there are two natural numbers n, m such that $n - m < N$,
for example $1, 2 \in N$ and $1 - 2 < N$

Integers Z are **closed** under–

Moreover Z is the **smallest set** containing N and closed under subtraction –

The set Z is called a **closure** of N under subtraction –

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Consider the directed graph R
& the directed graph R^*

Observe that

$$R^* = R \cup \{(a_i, a_i) : i = 1, 2, 3, 4\} \\ \cup \{(a_2, a_4)\}$$

Observe

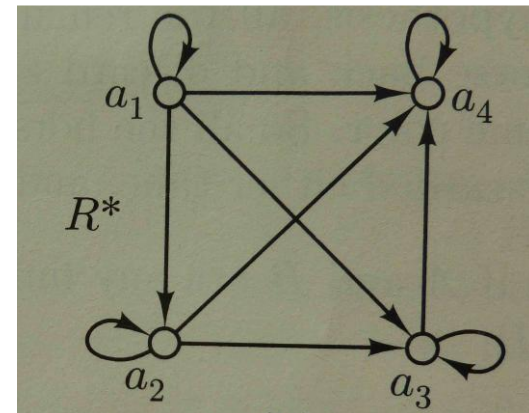
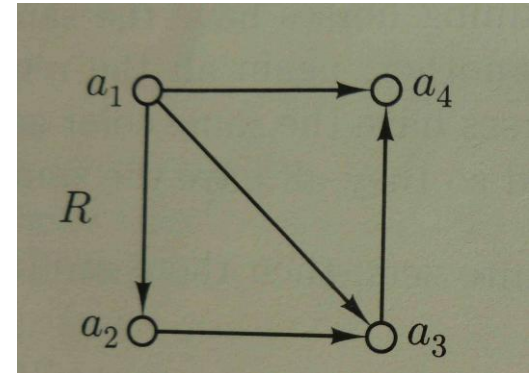
R is neither reflexive nor transitive

$$R \subseteq R^*$$

R^* is reflexive and transitive

Moreover R^* is also the **smallest set**
containing R that is **reflexive** and **transitive**

We call relation R^* the **reflexive, transitive closure** of R



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Two Definitions of R^*

1. R^* is called a **reflexive, transitive closure** of R iff
 $R \subseteq R^*$ and R^* is reflexive and transitive and is the smallest set with these properties
2. Let R be a binary relation on a set A
The **reflexive, transitive closure** of R is the relation
 $R^* = \{(a, b) \in A \times A : \text{there is a path from } a \text{ to } b \text{ in } R\}$

Definition 2 is a much **simpler definition** and algorithmically more interesting as it uses a simple notion of a path

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Definition 2 suggests the following algorithm for computing the reflexive transitive closure R^* of any given binary relation R over some finite set $A = \{a_1, a_2, \dots, a_n\}$

Algorithm 1

Initially $R^* := \phi$

for $i = 1, 2, \dots, n$ do

 for each i - tuple $(b_1, \dots, b_i) \in A^i$ do

 if $b_1, \dots, b_i$ is a **path** in R then add (b_1, b_i) to R^*

Running Time

$O(n^{n+1})$

Why?

In worst case n^n tuples to check.

Each tuple requires n operations

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Algorithm 2

Initially $R^* := R \cup \{(a_i, a_i) : a_i \in A\}$

While there are three elements a_i, a_j, a_k

if $(a_i, a_j); (a_j, a_k) \in R^*$ but $(a_i, a_k) \notin R^*$

then add (a_i, a_k) to R^*

This algorithm is more natural & intuitive than the first.

Running Time

$O(n^5)$

Why?

In worst case n^2 pairs to be added - iterations

In each iteration all triples needs to be searched (**Why**) – $O(n^3)$

Languages over Σ

Definition

Any finite set is called an **alphabet**

Elements of the **alphabet** are called symbols of the alphabet

Alphabet is any finite set of **symbols**

Alphabet Notation

We use a symbol Σ to denote the **alphabet**

$\Sigma = \{0, 1\}$ is called a **binary alphabet**

Remember

Σ can be ϕ as empty set is a **finite set**

We want to study $\Sigma \neq \phi$

Languages over Σ

We call finite sequences of the alphabet Σ as **words/strings** over Σ

We denote by e/λ the **empty string** over Σ

Let Σ be an **alphabet**.

We denote by Σ^* the set of **strings** over Σ

Observe that $e \in \Sigma^*$

So $\Sigma^* \neq \emptyset$

Languages over Σ

Language Definition

Given an alphabet Σ , any set L such that $L \subseteq \Sigma^*$ is called a *language over Σ*

Fact1:

For any alphabet Σ , any language over Σ is *countable*

Fact2:

For any alphabet $\Sigma \neq \phi$, there are *uncountably many* languages over Σ